Solving THERMAL MANAGEMENT Issues Is CRITICAL

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Selecting the right material, form and dimensions are key to producing a quality electronic product. Chris Nash, product manager for PCB Materials, Indium Corp., explains to Deepshikha Shukla how materials enable manufacturing of strong, reliable products that can endure inevitable physical shocks and thermal stresses associated with electronic devices, in applications ranging from Internet of Things (IoT) devices to next-generation, low-energy servers and automobile electronics

Q. What is the role of semiconductor materials in electronics?

A. Indium, germanium, gallium and tin metals and compounds are largely used in the manufacturing industry. These are used in transparent and conductive coatings in LCDs, grid storage batteries, LEDs, lasers, ultra-high-performance transistors, quantum dots, solar cells, catalysts, nuclear control rods and many more devices.

Q. Which is the largest-growing electronics sector? How do you cater to the needs of this segment?

A. Electronics/PCB assembly is one of the fastest-changing, most demanding markets on the planet. Industries are more focused on high electrical reliability, mechanical reliability and thermal reliability.

As electronics evolve to increasingly smaller, sophisticated and more powerful devices, we provide leading-edge materials and technology to solve assembly challenges of today and tomorrow.

Q. What is the role of flux in soldering?

A. A proper flux that can wet surfaces, clean oxides and still keep flip-chips in place during reflow is a basic requirement. A low- to medium-viscosity flux like flip-chip helps remove oxides during the reflow process from solder bumps or Cu-pillar micro-bumps. The most important function of flux is to clean the surface of the substrate to which it is soldered, to ensure proper wetting. The thermocompression bonding process uses ultra-low residue fluxes that can completely evaporate the extra flux and leave little residue behind.

Q. What does nanotechnology offer to thermal management materials?

A. Nanomaterials can be thinner than a piece of paper and, as with our NanoFoil product, can comprise more than thousand nano-layers of reactive metal. When NanoFoil is activated, it produces intense heat only in the area represented by the foil for approximately 1/1000th of a second. This precision allows it to accurately control timing, location and duration of heat energy. It is most suitable for situations such as a mismatch in coefficients of thermal expansion of two materials, temperature-sensitive materials, inability to use flux and ultra-rapid heating requirements.

Q. How can thermal management issues be solved?

A. Solving thermal management issues is critical. High-heat items require high-performance thermal interface materials to ensure reliable operation. This requires evaluating product designs, simulating operational conditions, measuring performance, and developing or selecting the right materials to solve problems. Thermal management materials enable electronic devices, such as laptops, cellphones, lighting systems and cell tower antennae, to operate at cooler temperatures, providing increased efficiencies, performance and longevity.

Q. What challenges can be solved with materials to achieve high performance?

A. The industry is worried about the amount of voiding in products, especially the automotive industry. Some applications require less than ten per cent voiding, which is challenging. However, we have met that challenge with Indium8.9HF solder paste, which delivers no-clean, halogen-free solutions to produce low voiding. It provides enhanced electrical reliability and improved stability during the printing process. This can also be coupled with high-reliability alloys to provide thermocycling reliability.

With transition from conventional vehicles to electric vehicles (EVs), requirement for electrical reliability and thermal reliability will increase. This will require thermal interface materials to suck out heat from the assembly and put it into a heat-sink and anticipate the heat very quickly. Our Heat-Springs solves these challenges in various components used in EVs.

Q. How do you achieve fully-characterised materials and processes in leading-edge technology applications?

A. We provide technical support to our customers for material evaluations and process upgrades. We also help develop custom solutions for their technical problems and optimise their processes to increase yields and revenues, and reduce defects. We supply materials to global electronics, thin-film, semiconductor, thermal management and solar markets. We practice responsible hazardous substance control and management methods for reuse, recycling and proper disposal of these materials.

Our products are compliant with laws and regulations, including solders and fluxes, brazes, thermal interface materials, sputtering targets, indium, gallium, germanium and tin metals, inorganic compounds, and NanoFoil. We also provide high-quality, exceptional solder products designed to seamlessly integrate for increased overall reliability in automotive and cellphones. EFY